

Anna Fisher Lopez

Data & ML Engineer

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PROFESSIONAL EXPERIENCE

Data Engineering Summer Intern

Blue Forest

Jun 2026 – Present

San Francisco, CA

- Built a data pipeline (Python, DuckDB, Geoparquet) to visualize and quantify environmental and economic impacts of wildfires on water resources in the Sierra Nevadas, for internal and external use.
- Collecting, cleaning, and organizing large amounts of data from a wide range of sources for deployment through Streamlit.

Teaching Assistant for CS 109: Probability for Computer Scientists

Stanford University, Dept. of Computer Science

Sep 2025 – Jun 2026

Stanford, CA

- Selected as one of 7 TAs from a highly competitive applicant pool
- Taught 2-4 interactive sections per week and led 4 hours of office hours covering applied statistics and probability.

AI/ML Research Assistant

Stanford University, Dept. of Computer Science & School of Sustainability

Feb 2025 – Dec 2025

Stanford, CA

- Indep. project guided by Andreas Paepcke: Analyzed bat communication patterns using graph-based neural networks, demonstrating identifiable patterns in bat chirp sequences
- Sheshadri Lab: Calculated momentum fluxes and developed a UNet model using ERA5 data to build better climate models

Strategy and Software Intern

Gemini Energy Solutions

Jun 2024 – Aug 2024

Washington, D.C.

- Improved usability and reliability of Gemini's energy audit assistance platform through backend software updates (TypeScript)
- Conducted a nationwide market priority study to identify priority areas for Gemini's new projects
- Coordinated with clean energy contractors in different states to align project execution with energy-efficiency and budget goals

EDUCATION

Stanford University

M.S. Computer Science (AI Track)

Sep 2025 – Jun 2026

- Relevant Courses & Projects: Deep Learning for Computer Vision, Reinforcement Learning, Machine Learning with Graphs, Stanford Climate Ventures

Stanford University

B.S. Physics (Physical Oceanography Track)

Sep 2021 – Jun 2025

- Other research areas: Astrophysics, Equity in Energy Efficiency

SKILLS & INTERESTS

Skills — Python, C, C++, PyTorch, TensorFlow, Climate Modeling, ArcGIS, Probability/Statistics, Spanish (spoken & written working proficiency)

Domain Interests — Renewable Energy, Sustainability & AI, Grid Optimization, Climate Change Mitigation Tech & Strategy

ML RESEARCH

Attention UNets for Climate Modeling

Research, Sheshadri Lab

2025 | Stanford, CA

- Cleaned and validated massive amounts of ERA5 data
- Trained a UNet model on calculated momentum fluxes using cleaned ERA5 data, adding 3D gravity wave modeling to existing climate models.

Material-Based Perception & Image Segmentation

Deep Learning for Computer Vision (CS 231N)

2026 | Stanford, CA

- Developed a material-based semantic image segmentation pipeline, reducing inference times by 50% over SotA with comparable semantic identification
- Trained a ResNet-50 classifier using AWS EC2 with experiment tracking and checkpointing through Weights & Biases.

GNNS for Acoustic Pattern Recognition

Machine Learning with Graphs (CS 224W)

2026 | Stanford, CA

- Analyzed bat communication patterns using graph-based neural networks, demonstrating identifiable patterns in bat chirp sequences.